Demonstration of Improvement in On-Resistance Versus Breakdown Voltage Trade-Off for Top-Gate High-Voltage InGaO TFTs

Jiayao Hao, Huibo Liu, Peiyan Hong, Dongying Qiu, Zhenyu Zhang, and Xuefei Li

***Abstract*— In this work, we report a method to improve the trade-off between the breakdown voltage (BV) and specific on-resistance ($R_{ON, sp}$) of top-gate amorphous indium gallium oxide (IGO) high-voltage thin-film transistors (HV TFTs) with selective F plasma treatment for offset regions. X-ray photoelectron spectroscopy (XPS) confirmed that F plasma treatment effectively increased the oxygen vacancy (V_O) concentration in the IGO film, improving the conductivity of the offset regions. Accordingly, the offset-region resistance of top-gate HV IGO TFTs is reduced by 58.3%, while its pristine threshold voltage (V_{TH}) of -2.2 V and subthreshold swing (SS) of 81 mV/dec are preserved. Specifically, the top-gate HV IGO TFT with L_{GD} = 10 μm achieves a high BV of 1015 V and an $R_{ON, sp}$ of 79 mΩ·cm^2. Our results show the great potential of top-gate HV IGO TFTs in high-voltage and high-power applications.**

***Index Terms*— Indium gallium oxide (IGO), top-gate, F plasma, high-voltage thin-film transistors (HV TFTs).**

I. Introduction

HIGH-VOLTAGE thin-film transistors based on amorphous oxide semiconductors (AOSs) have emerged as compelling contenders for high-power devices because of their wide bandgap (>3 eV) and high carrier mobility (>30 cm^2/V·s) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. Currently, the BV of bottom-gate HV TFTs is typically constrained to below 1000 V. Although the electric field distribution in the offset region is relatively uniform, TCAD simulations indicate that under the same L_{GD} and drain voltage (V_D), top-gate devices feature a lower peak electric field than their bottom-gate counterparts do [Fig. 1(a) and (b)], and such a peak constitutes the main cause of device breakdown [12]. This implies that a larger L_{GD} is required for bottom-gate HV TFTs to achieve an equivalent breakdown voltage. Moreover, when the gate length (L_G) is approximately equal to the gate-to-source length (L_{GS}), the peak electric fields at the gate stack edge region are minimized, which effectively suppresses the electric field concentration at the gate stack edge and thus improves the BV of TFTs [12]. Although the offset region separates the gate and drain electrodes to alleviate the electric field crowding at the gate–drain edge, they introduce considerable series resistance, degrading the on-state current and creating an inherent trade-off between the BV and on-resistance. To reduce offset-region resistance in bottom-gate devices, two typical practices have been reported previously: selective doping of the offset regions with plasma or the application of a low-resistance ITO capping layer to the channel offset regions [13], [14], [15]. While these methods address offset-region resistance optimization issues for bottom-gate transistors, a practical strategy tailored to reduce such resistance in top-gate HV TFTs is still lacking.

In this work, we propose a selective F plasma treatment strategy to decrease the $R_{ON,sp}$ of top-gate HV IGO TFTs. These devices exhibit a significant increase in the on-state

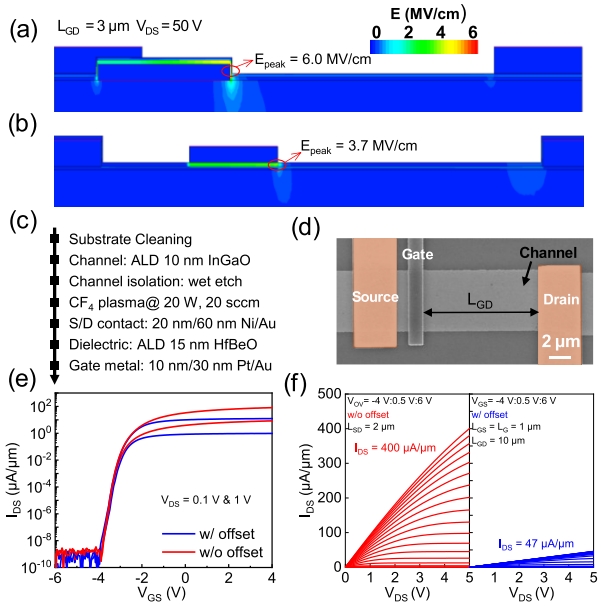


Fig. 1. TCAD-simulated electric field intensities of the (a) bottom-gate and (b) top-gate structures. (c) Fabrication process flow and (d) SEM image of the top-gate IGO HV TFTs. (e) Transfer and (f) output characteristics of top-gate IGO HV TFTs with and without an offset structure.

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current (I_{ON}) compared with devices without F plasma treatment. The BVs of the devices exhibit a strong linear relationship with the gate-to-drain length (L_{GD}), with an average breakdown field strength (BV/L_{GD}) of 0.98 MV/cm. The top-gate HV IGO TFT with L_{GD} = 10 μm achieves a high BV of 1015 V. The proposed top-gate HV IGO TFTs present an outstanding trade-off relationship between the V_{BD} and $R_{ON,sp}$, surpassing the theoretical limit of Si in the BFOM plot.

II. Device Fabrication

The fabrication process and top-view scanning electron microscopy (SEM) image of the top-gate IGO HV TFTs are shown in Fig. 1(c) and (d), respectively. After the sapphire substrate was cleaned, a 10-nm amorphous IGO channel layer was grown by ALD at 225 °C. The ALD process consists of separate In_2O_3 and Ga_2O_3 subcycles: for the In_2O_3 subcycles, trimethylindium (TMIn) and water (H_2O) were selected as the indium and oxygen precursors, whereas trimethylgallium (TMGa) and ozone were chosen for the Ga and O sources in the Ga_2O_3 subcycles. The channel region was defined via electron beam lithography (EBL) followed by wet etching with 1:2 hydrochloric acid. The offset regions of the HV IGO TFTs were subsequently selectively treated in a reactive ion etching (RIE) system using a CF_4 plasma with a power of 20 W and a flow rate of 20 sccm. A 15-nm-thick HfBeO (14:1) top-gate dielectric was deposited at 250 °C after deposition of 20 nm/60 nm Ni/Au as S/D. Finally, 10 nm/30 nm Pt/Au was deposited as the gate metal. The entire fabrication process temperature is as low as 250 °C. Electrical measurements of the IGO HV TFTs were carried out via Agilent semiconductor parameter analyzers B1500A (low-voltage) and B1505A (high-voltage). The fabricated IGO HV TFTs feature L_G of 1 μm, L_{GS} of 1 μm and L_{GD} of 3, 6, and 10 μm for different BVs.

III. Results and Discussion

Fig. 1(e) and (f) show the transfer and output characteristics of top-gate IGO HV TFTs with and without offset structures. Compared with the device without an offset region, the I_{ON} at V_{GS} = 6 V and V_{DS} = 5 V of the device with an offset structure decreases from 400 μA/μm to 47 μA/μm. Fig. 2(a) shows atomic force microscopy (AFM) images of IGO films without and with F plasma treatment, which exhibit root-mean-square (RMS) surface roughness values of 0.21 nm and 0.14 nm, respectively. As shown in Fig. 2(b), the extracted bandgap values for both samples are nearly identical at 3.53 eV, confirming that F plasma treatment induces no obvious change in the bandgap (E_g) of the IGO thin film. Under the assumption that the material exhibits avalanche breakdown behavior [16], [17], the critical breakdown electric field (E_{CR}) of the IGO films is estimated to be 4.09 MV/cm.

To characterize the static I-V characteristics of the top-gate HV IGO TFTs, the contact resistance (R_C) and sheet resistance (R_{SH}) of these two groups of devices with HfBeO as the top-gate dielectric were subsequently measured via the transfer length method (TLM), as shown in Fig. 2(c). The R_{SH} values were 10.5 kΩ/sq and 4.55 kΩ/sq, respectively, while the R_C were 840 Ω · μm and 750 Ω · μm, respectively. These results indicate that F plasma treatment effectively reduces both the R_{SH} of the IGO film and the R_C of the fabricated devices.

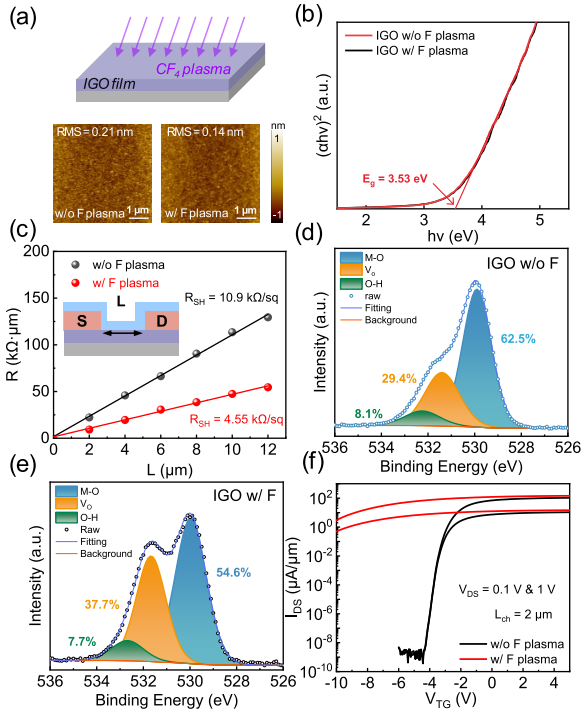


Fig. 2. (a) Schematic of the ALD-grown IGO film treated with F plasma and AFM measurement of the surface roughness of the IGO film without and with F plasma. (b) $(\alpha hv)^2$-hv plot for IGO films without and with F plasma. (c) TLM to extract the sheet resistance of IGO films without and with F plasma under a HfBeO dielectric. (d) and (e) XPS spectra of deconvoluted O 1s peaks for IGO films without and with F plasma. (f) Transfer characteristics of top-gate IGO TFTs without and with F plasma.

XPS was performed as shown in Fig. 2(d) and (e). After F plasma treatment, the concentration of V_O in the IGO film increased from 29.4% to 37.7%, which is the key reason for the increased electrical conductivity of the treated IGO film [18], [19]. While F plasma treatment effectively reduces the R_{SH} of the IGO channel, as shown in Fig. 2(f), it induces a significant negative shift in the threshold voltage. Notably, the threshold voltage of IGO HV TFTs is dominated by the gate-controlled channel region. To address this issue, we propose a selective F plasma treatment strategy applied exclusively to the gate-source and gate-drain offset regions to reduce the access resistance of these regions.

As illustrated in Fig. 3(a), the offset regions are first defined via EBL, followed by the selective application of F plasma treatment exclusively to the prepatterned offset regions. This approach ensures that F plasma treatment is precisely confined to the target offset regions to reduce their resistance, leaving the gate-controlled channel region unaffected. The transfer and output characteristics of top-gate IGO HV TFTs (L_{GD} = 6 μm) without and with offset plasma treatment are shown in Fig. 3(b)–(d), respectively. As expected, the threshold voltage (V_{TH} = -2.2 V) and the subthreshold swing (SS = 81 mV/dec) of the devices did not significantly change, whereas the on-state current notably increased from 108 μA/μm to 223 μA/μm. The measured I_{ON} at V_{GS} = 6 V and V_{DS} = 10 V and the corresponding R_{ON} of the top-gate IGO HV TFTs as a function of the L_{GD} are shown in Fig. 3(e). An increase in L_{GD} results in a decrease in I_{ON} and an increase in R_{ON}. The experimental and simulation results of (R_S + R_D)/R_{ON} as a function of L_{GS} + L_{GD} for devices without and with offset

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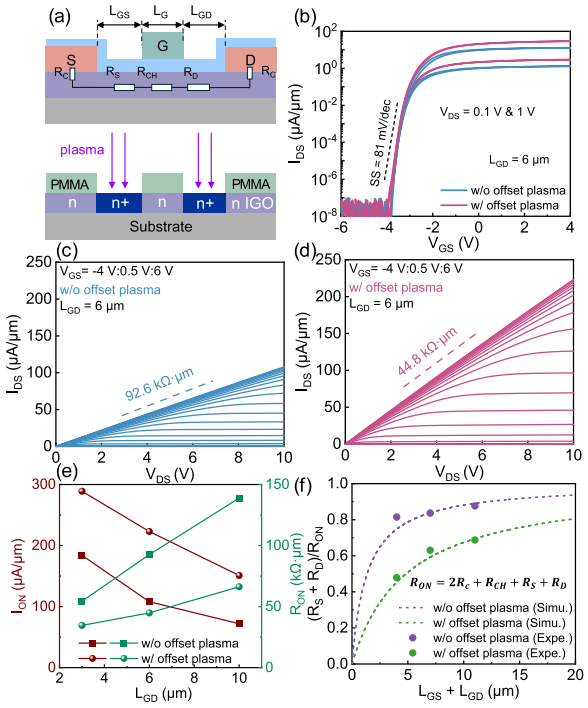


Fig. 3. (a) Schematic of selective F plasma treatment on offset regions of top-gate HV IGO TFTs. (b) Transfer characteristics of top-gate IGO HV TFTs without and with offset plasma treatment. Output characteristics of top-gate HV IGO TFTs (c) without and (d) with offset plasma treatment. (e) Measured I_{ON} and extracted R_{ON} of top-gate IGO HV TFTs without and with offset plasma treatment at V_{GS} = 6 V and V_{DS} = 10 V. (f) Experimental and simulation results of $(R_S + R_D)/R_{total}$ as a function of $L_{GS} + L_{GD}$ for devices without and with offset plasma treatment.

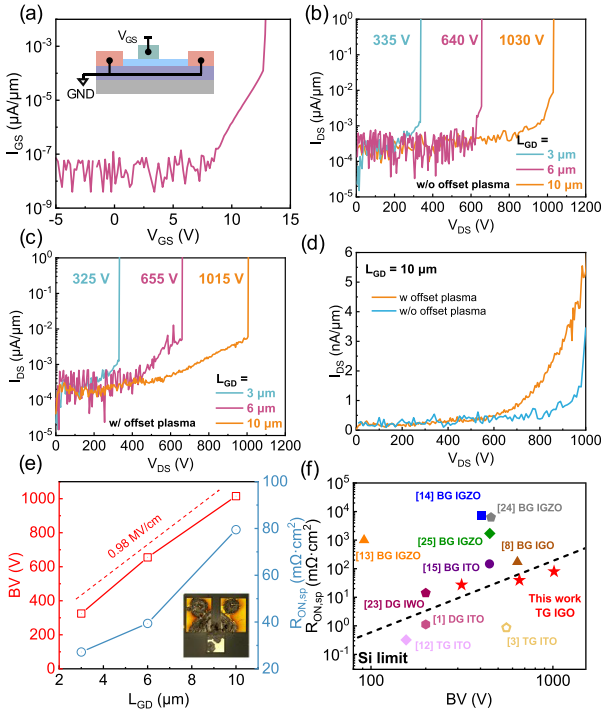


Fig. 4. (a) Gate leakage and breakdown characteristics of top-gate HV IGO TFTs. OFF-state I–V characteristics of top-gate IGO HV TFTs for L_{GD} = 3, 6, and 10 μm: (b) without offset plasma treatment and (c) with offset plasma treatment. (d) Linear-scale OFF-state I–V characteristics of top-gate high-voltage IGO TFTs with L_{GD} = 10 μm. (e) BV and $R_{ON,\,sp}$ as a function of L_{GD}. (f) Benchmark of $R_{ON,\,sp}$ vs BV of recently reported AOS-based HV TFTs.

plasma treatment are shown in Fig. 3 (R_S and R_D are the source and drain access resistances, respectively). Selective F plasma treatment of the offset regions of top-gate HV IGO TFTs effectively reduces the contribution of the offset-region resistance ($R_S + R_D$) to the total R_{ON}.

The gate hard BV of the top-gate HV IGO TFTs is 12.7 V, with the source and drain grounded, as shown in Fig. 4(a). The OFF-state I-V characteristics were measured in 3M FC-40 fluorinated oil to prevent air breakdown prior to the transistor [20]. As shown in Fig. 4(b) and (c), the BVs of the top-gate HV IGO TFTs without and with offset plasma treatment corresponding to L_{GD} of 3 μm, 6 μm and 10 μm were 335 V, 640 V, 1030 V, 325 V, 655 V and 1015 V, respectively. The BVs of IGO HV TFTs with and without offset plasma treatment are nearly identical (differences < 3%) since the bandgap and E_{CR} of IGO thin films show no obvious changes after plasma treatment, making the BVs nearly independent of offset plasma treatment. As shown in Fig. 4(d), compared with devices without offset plasma treatment, devices with offset plasma treatment have a greater off-state leakage current, which can be attributed to the higher oxygen vacancy defect densities in the offset region induced by plasma treatment, indicating that off-state leakage is likely dominated by trap-assisted thermionic field emission (TA-TFE) [21], [22] in the IGO channel layer. Fig. 4(e) shows the BV at V_{GS} = −5 V and $R_{ON,sp}$(defined as $R_{ON,sp} = R_{ON} \times W \times L$) of the top-gate IGO HV TFTs as a function of L_{GD}, where a higher BV is achieved at the expense of an increased $R_{ON,sp}$ as the L_{GD} increases. Notably, as shown in Fig. 4(e), the device is damaged and cannot be recovered after the breakdown measurement. There is a strong linear relationship between BV and L_{GD}, and the average breakdown field (BV/L_{GD}) of top-gate IGO HV TFTs is ~0.98 MV/cm, which is significantly greater than that of our previously reported bottom-gate HV IGO TFTs (~0.35 MV/cm) [8]. The linear BV-L_{GD} relationship originates from the uniform electric field in the gate-drain offset region, whereas local electric field crowding near the gate electrode triggers device breakdown and results in the average breakdown field being lower than the estimated theoretical E_{CR} for the IGO material.

The benchmarks of $R_{ON,sp}$ versus BV of recently reported AOS-based HV TFTs are shown in Fig. 4(f) [8], [12], [13], [14], [15], [23], [24], [25]. The top-gate IGO HV TFTs with L_{GD} = 10 μm in this work achieve a record BV of 1015 V–the highest reported for AOS-based top-gate HV TFTs. The proposed top-gate IGO HV TFTs with plasma treatment surpassed the Si limit in the BFOM plot [26].

IV. Conclusion

In summary, we fabricated top-gate HV IGO TFTs and selectively performed F plasma treatment on the offset regions of the devices, which increased the oxygen vacancy concentration in the offset regions and effectively reduced their resistance. The BVs of the devices are strongly linearly correlated with the L_{GD}, with an average breakdown field strength of 0.98 MV/cm. The top-gate HV IGO TFT with L_{GD} = 10 μm has a high BV of 1015 V and an $R_{ON,sp}$ of 79 mΩ·cm^2, effectively improving the tradeoff between $R_{ON,sp}$ and BV. These results highlight the significant potential of top-gate HV IGO TFTs in low-temperature power electronics applications.

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